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Evaluating prodrug characteristics of a novel anticoagulant fusion protein neorudin, a prodrug targeting release of hirudin variant 2-Lys47 at the thrombosis site, by means of in vitro pharmacokinetics

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ABSTRACT

Recombinant neorudin (EPR-hirudin, EH), a low-bleeding anticoagulant fusion protein, is an inactive prodrug designed to be converted to the active metabolite, hirudin variant 2-Lys47 (HV2), locally at the thrombus site by FXa and/or FXIa, following activation of the coagulation system. Our aim was to evaluate the prodrug characteristics of EH by comparing the biotransformation of EH and HV2 in biological matrices, including rat blood, liver, and kidney homogenates, demonstrating the cleavage of EH to HV2 by FXa and FXIa, and comparing the conversion of EH to HV2 between fresh whole blood and whole-blood clot homogenate, using ultra-performance liquid chromatography-mass spectrometry (UPLC-MS/MS). Both EH and HV2 were stable in blood and unstable in the liver and kidney homogenates. Eight EH metabolites and eight HV2 metabolites identified as N-terminal fragments were found in the liver and kidney. C-terminal proteolysis is therefore the major metabolic pathway, with serine/cysteine carboxypeptidases and metallocarboxypeptidases being responsible for the degradation of EH and HV2 in the liver and kidney, respectively. EH was cleaved to release HV2 by FXIa. Higher levels of HV2 were produced from EH in the whole-blood clot homogenate, in which the coagulation system was activated compared with those in fresh whole blood. In conclusion, the metabolism of EH and HV2 shares the same cleavage pattern, and EH is transformed into HV2 when the coagulation system is activated, where FXIa is a specific enzyme. Our in vitro study revealed the anticipated prodrug characteristics of EH newly designed as an inactive prodrug of hirudin.

1. Introduction

Hirudin, a single-chain polypeptide of approximately 7 kDa consisting of 65 amino acids, was first extracted from the leech salivary gland secretions of medicinal by Markwardt in 1955 (Markwardt, 1955; Dodt et al., 1984). Hirudin is a powerful natural thrombin inhibitor, with Ki and Kd at the pico- and femtomolar range, respectively (Lyer and Fareed, 1996; Huntington, 2014). Three variants of hirudin have been identified: hirudin-VV (hirudin variant 1), hirudin-IT (hirudin variant 2), and hirudin-PA (hirudin variant 3); these exhibit comparable activities in standard coagulation assays (Müller et al., 2016). Hirudin has an asymmetric structure, consisting of a compact, hydrophobic amino-terminal core region with alternating polar and non-polar segments, three disulphide bonds, and an extended and highly acidic C-terminal region (Johnson et al., 1989). Hirudin inhibits thrombin activity by binding at a 1:1 stoichiometric ratio (Markwardt and Walsmann, 1958). The C-terminal, hydrophobic core, and N-terminal regions of hirudin bind to the anion-binding (fibrinogen binding) exosite, active site, and the apolar binding regions, respectively, of thrombin. Upon hirudin binding, a loop of 10 amino acid residues from the catalytic site is displaced, resulting in loss of thrombin catalytic activity (Johnson, 1994). C-terminal modifications of pharmacologically active peptides frequently result in alteration of receptor potency, selectivity, activation, or pharmacokinetic profile (Krstenansky et al., 1989).

Hirudin-based anticoagulants include recombinant hirudin (lepirudin and desirudin) and hirudin analogues (bivalirudin). Lepirudin was approved for the treatment of heparin-associated thrombocytopenia

Abbreviations: EH, EPR-hirudin, Neorudin; HV2, hirudin variant 2-Lys47; DVT, deep vein thromboembolism; NEM, *N*-ethylmaleimide; CVDs, cardiovascular diseases

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(HIT) and associated thromboembolic diseases by the FDA on 6 March 1998, but was withdrawn for commercial reasons in 2012. Desirudin was approved for preventing deep vein thromboembolism (DVT) following hip or knee arthroplasty by the FDA on 4 April 2003. Bivalirudin was approved by the FDA on 15 December 2000, for treatment of unstable angina, HIT, or elevated HIT risk following percutaneous coronary intervention (PCI). However, the use of these thrombin inhibitors has several drawbacks, such as bleeding, strong dependence on renal function, lack of antidote, and rebound hypercoagulability, particularly when associated with thrombolytic agents and acetylsalicylic acid. Controlling the dose, periodic monitoring, and strict selection of patients are therefore applied to overcome these problems (Greinacher and Warkentin, 2008). There is a pressing need for the development of a highly effective and safe anticoagulant hirudin derivative.

Neorudin (EH) is a novel anticoagulant fusion protein with a molecular weight of 7.284 kDa, recombinant expressed in *Escherichia coli* (Mao et al., 2016; Wu et al., 2016). EH was developed as a prodrug that releases hirudin variant 2-Lys47 (HV2) only at the thrombosis site to treat DVT with low bleeding risk. The antithrombotic efficacy and safety of EH were assessed via in-vivo and in-vitro experiments (Wang et al., 2013). The EPR (Glu, Pro, and Arg) oligopeptide that is recognised by FXa and/or FXIa was introduced at the N-terminus of HV2, affecting the inhibitory effects of hirudin on thrombin. If EH is not cleaved, the hirudin N-terminal is blocked and its anticoagulant activity is lost. Activation of the coagulation system leads to enrichment of coagulation factors at the thrombosis site, releasing HV2 from EH by cleavage of EPR by FXa and/or FXIa, resulting in a strong anticoagulant effect without systemic bleeding and thus a low bleeding risk (Peter et al., 2003; Zhang et al., 2008). The specificity of EH cleavage by FXa and FXIa has been confirmed by functional assays in vitro (Zhang et al., 2008); however, biotransformation of EH and HV2 in biological matrices and the bioconversion of EH into HV2 at the clot site by in vitro pharmacokinetic assay have not yet been reported.

In this work, we aimed to evaluate prodrug characteristics of EH by means of in vitro pharmacokinetic experiments including comparing the biotransformation of EH and HV2 in biological matrices, including rat blood, liver, and kidney homogenates, demonstrating the cleavage of EH to HV2 by factor Xa and XIa, and comparing the conversion of EH to HV2 between fresh whole blood and whole-blood clot homogenates. We report, for the first time, the metabolic characteristics of EH and its active metabolite, HV2. The corresponding metabolites and enzymes involved were identified. In addition, we provide quantitative and qualitative evidence that EH is capable of transforming to HV2 by FXIa and in clot homogenates. Furthermore, the biotransformation exploration of EH and HV2 provides directions for efforts aimed at the biotransformation of mid-class peptides and small proteins.

2. Materials and methods

2.1. Materials

EH (MW 7284.2 Da, purity 97%) and HV2 (MW 6906.5 Da, purity 98%) were synthesised at the Institute of Radiation Medicine (Beijing, China), and the amino acid sequences are shown in Fig. 1. DL-thiorphan, *N*-Ethylmaleimide (NEM), bestatin, phosphoramidon, trisodium citrate dehydrate, and Krebs-Henseleit buffer (KHB; with 0.2% D-glucose, pH 7.4) were purchased from Sigma-Aldrich (St. Louis, MO, USA). FXa and FXIa were obtained from Haematologic Technologies, Inc. (Essex Junction, VT, USA). Ultra-pure water was obtained using a Milli-Q Advantage A10 (Millipore, Billerica, MA, USA). Acetonitrile, methanol, formic acid, and isopropanol (chromatographic grade) were obtained from Thermo Fisher Scientific (Waltham, MA, USA). BD Vacutainer Plus Serum and SST™ tubes were purchased from BD Bioscience (San Jose, CA, USA).

2.2. Methods

2.2.1. Preparation of biological matrices

Preparation of liver and kidney homogenates Six male Wistar rats, weighing 220–240 g, were purchased from the Academy of Military Medical Sciences (Beijing, China). Animal experiments were approved by the Animal Care and Welfare Committee of the Institute of Radiation Medicine, Academy of Military Medical Sciences (Beijing, China). Animals were fasted overnight with water ad libitum and anaesthetised using isoflurane inhalation anaesthesia (Matrix, Midmark, USA). After quickly opening the abdominal cavity, the kidney and the liver were perfused with cold 0.9% saline via the portal and inferior caval veins until the blood was rinsed off. Kidneys and livers were excised and washed with cold 0.9% saline, followed by trimming of fat tissues and cleaning with filter paper. Organs were weighed, cut into small pieces, transferred into cold KHB at a ratio of 1: 4 (gram of organ/ml of KHB), and homogenised using an IKA T25 digital ULTRA TURRAX at 14000 rpm for 1 min on crushed ice. Homogenates were centrifuged at 1000 ×*g* for 5 min at 4 °C, and the protein concentration of supernatants was determined using Pierce BCA Protein Assay Kit (Thermo Fisher Scientific, Waltham, MA, USA). Supernatants were stored at −80 °C until use.

Preparation of fresh whole blood and whole-blood clot homogenate Wistar rats weighing 220–240 g were anaesthetised and blood was collected from the inferior caval vein. Blood was placed in 3.8% sodium citrate-containing tubes (volume ratio 9: 1) and inverted 3–4 times for thorough mixing. Whole-blood clot homogenate was prepared using BD SST™ tube blood collection. The tubes were inverted 5 times, allowed 30 min of clotting time, and then homogenised using an IKA T25 digital ULTRA TURRAX at 7000 rpm for 1 min on crushed ice.

2.2.2. Comparative studies of EH and HV2 metabolism

Metabolic stability in fresh whole blood, liver, and kidney homogenates EH or HV2 peptide stock solution (5 µL) and 500 µL of individually prepared liver (13.6 mg/mL), kidney (16.1 mg/mL), and fresh whole blood were mixed at 37 °C with shaking. Final concentration of peptides was 20 µg/mL. After 0 min, 5 min, 15 min, 30 min, 1 h, 2 h, and 3 h, aliquots (50 µL) were removed and the peptides were precipitated with 200 µL methanol. After centrifugation at 4415 ×*g* for

Neorudin (EH): Glu-Pro-Arg-HV2

Hirudin variant-2-Lys47 (HV2):

Ile-Thr-Tyr-Thr-Asp-Cys-Thr-Glu-Ser-Gly-Gln-Asn-Leu-Cys-Leu
Cys-Glu-Gly-Ser-Asn-Val-Cys-Gly-Lys-Gly-Asn-Lys-Cys-Ile-Leu
Gly-Ser-Asn-Gly-Lys-Gly-Asn-Gln-Cys-Val-Thr-Gly-Glu-Gly-Thr
Pro-Lys-Pro-Glu-Ser-His-Asn-Asn-Gly-Asp-Phe-Glu-Glu-Ile-Pro
Glu-Glu-Tyr-Leu-Gln

Fig. 1. The amino acid sequences of neorudin (EH) and hirudin variant-2-Lys47 (HV2).

30 min at 4 °C, the supernatant of samples was analysed using ultra-performance liquid chromatography-mass spectrometry (UPLC-MS/MS) for quantification.

Metabolite identification in liver and kidney homogenates EH or HV2 peptide stock solution (5 μL) and 500 μL of individually prepared liver and kidney were mixed at 37 °C with shaking. The final concentration of peptides was 200 μg/mL. After 0 min and 5 min for kidney or 3 h for liver, aliquots (50 μL) were removed and the peptides were precipitated with 200 μL methanol. The suitable incubation time were designed based on metabolic stability of EH and HV2. After centrifugation at 4415 ×*g* for 30 min at 4 °C, the supernatant was diluted twice using ultra-pure water before UPLC-MS/MS analysis.

Inhibition of EH and HV2 metabolism in liver and kidney homogenates In this assay, serine peptidase inhibitor AEBSF (2 mM), cysteine peptidase inhibitor *N*-ethylmaleimide (4 mM), metallopeptidase inhibitor DL-thiorphan (2 mM), endopeptidase inhibitor phosphoramidon (0.2 mM), and aminopeptidase inhibitor bestatin (0.7 mM) (Rawlings, 2010; Rawlings et al., 2011) were added to liver and kidney homogenates, vortexed, and pre-incubated for 10 min (liver) or 30 min (kidney) at 37 °C. Samples were incubated with EH or HV2 (200 μg/mL) for an additional 5 min (kidney) or 3 h (liver) at 37 °C. The reaction was stopped with the addition of methanol, and the samples were analysed as described above for qualitative and semi-quantitative analyses.

2.2.3. In vitro conversion of EH to HV2

The cleavage of EH to release HV2 by FXa and FXIa In this typical reaction system, 75 μg of EH (0.2 mg/mL) was cleaved at 37 °C for a certain period of time by 7.5 μg of FXa (7.5 mg/mL) and FXIa (2.3 mg/mL), respectively, in 20 mM Tris-HCL, 150 mM NaCl, and 2 mM $CaCl_2$ (PH 7.5). After 0 min, 1 h, 2 h, 3 h, 5 h, 9 h and 24 h, aliquots (50 μL) were taken and stopped with 200 μL methanol. After centrifugation at 4415 ×*g* for 30 min at 4 °C, the supernatant of samples was analysed using UPLC-MS/MS for quantification.

Release of HV2 from EH in fresh whole blood and whole-blood clot homogenate In this experiment, the release of HV2 from EH in fresh whole blood and whole-blood clot homogenate was compared. The whole-blood clot homogenates were incubated either with or without the addition of EH (200 μg/mL) at 37 °C. The same procedures were applied to fresh whole blood. After 0 min and 3 h, aliquots (50 μL) were taken and precipitated with 200 μL methanol. After centrifugation at 4415 ×*g* for 30 min at 4 °C, the supernatant was diluted twice using ultra-pure water before UPLC-MS/MS for qualitative and semi-quantitative analyses.

2.2.4. LC-MS analysis

The quantification of samples was according to our previously reported method (Dong et al., 2017). The analytical range in different tissues was from 50 to 50,000 and from 10 to 10,000 ng/mL for EH and HV2, respectively, and the calibration curves were fitted with a linear regression curve ($1/X^2$).

The qualitative and semi-quantitative analyses of samples were conducted with a Dionex UltiMate 3000 UHPLC focused and Q Extractive Focus combined system (Thermo Fisher, Milford, MA, USA) with an electrospray ionization (ESI) source. Waters ACQUITY UPLC BEH300 C18 (300 Å, 2.1 mm × 50 mm, 1.7-μm) was used to perform separations. The mobile phase consisted of 2 parts, including A, which contained a 0.1% volume ratio of formic acid in water, and B, which contained a 0.1% volume ratio of that in acetonitrile. Chromatography was performed at a flow rate of 0.5 mL/min using a multi-step gradient with a mobile phase B at 2% for 3 min, at 2 to 80% for 13 min, at 80% for 1 min, then back to the initial gradient at 17.01 min, with total assay duration of 25 min. The injection volume was 5 μL. MS was performed on a Q Extractive Focus mass spectrometer with an ESI in positive polarity. Full MS scan range of 200–3000 *m/z* was acquired in the Q Extractive Focus with a 70,000 resolution. The lock mass option was set as best and chromatogram peak width set as 6 s. The dd-MS_2 discovery was performed by selectively targeting the *m/z* values of the compounds of interest, with an isolation width of 3 Da, a CE of 23 and 26 for EH and HV2, respectively, and a resolution of 70,000. ESI source conditions were as follows: sheath gas flow rate, 40 psi; aux gas flow rate, 10 arbitrary units; spray voltage, 3.5 kV; capillary temperature, 325 °C; S-lens RF level, 55 AU; aux gas heater temperature, 300 °C. Data acquisition was performed using Xcalibur software version 4.0.27.13 (Thermo Fisher Scientific).

2.3. Data analysis

The terminal rate constant (ke) of EH and HV2 in tissue homogenates during in vitro incubation was obtained using log-linear regression analysis of substrate remaining percent (%)-time points. The half-life ($T_{1/2}$) was calculated according to the following equation: $T_{1/2} = 0.693/ke$.

The ProteinProspector tool version 5.19.04 (University of California, San Francisco, CA) was used to deduce the amino acid sequence.

3. Results

3.1. Comparative studies of EH and HV2 metabolism

3.1.1. Metabolic stability in fresh whole blood, liver, and kidney homogenates

The metabolic stability of EH and HV2 in rat blood, liver, and kidney homogenates was determined, and is shown in Fig. 2. The $T_{1/2}$

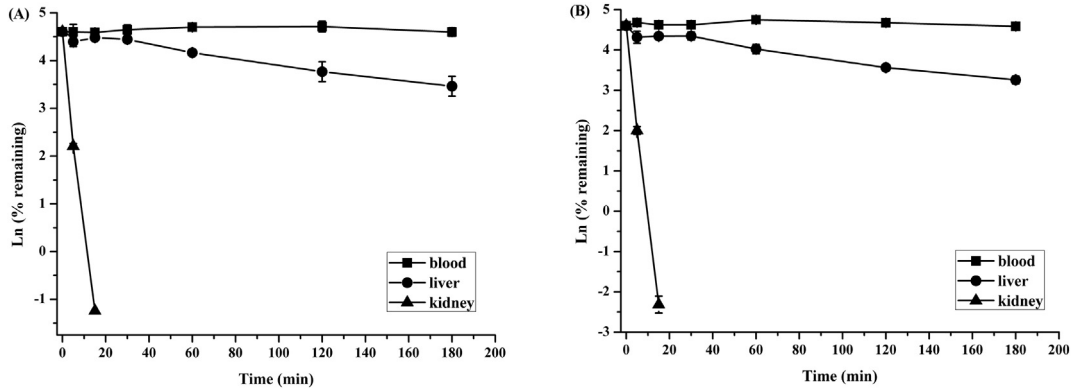


Fig. 2. Time course of EH and HV2 degradation upon incubation in rat blood, liver, and kidney homogenates at 37 °C. The values on the y axis represent Ln (% remaining) and the remaining % is the percentage recovery versus T_{0min}. Data represent mean ± SD from three replicate incubations. (A) EH degradation in rat blood, liver, and kidney homogenates. (B) HV2 degradation in rat blood, liver and kidney homogenates.

Table 1
In vitro stability of EH and HV2 in rat blood, liver, and kidney homogenates.

Compound	Matrices	$T_{1/2}$[a] (min)
EH	Blood	ND[b]
	Liver	116 ± 25
	Kidney	1.70 ± 0.19
HV2	Blood	ND
	Liver	98.4 ± 10.7
	Kidney	1.52 ± 0.05

[a] $T_{1/2}$ values represent mean ± SD from three replicate incubations.
[b] ND, not determined.

of the proteins is summarised in Table 1. The degradation rate of EH and HV2 in the different tissues was, in decreasing order: kidney > liver > blood. No significant degradation was observed in blood after 3 h of incubation. In contrast, EH and HV2 were almost completely degraded in the first 5 min of incubation (< 10% remaining) in kidney homogenates. The $T_{1/2}$ of EH and HV2 in liver homogenates was 116 and 98.4 min, respectively.

3.1.2. Metabolite identification in liver and kidney homogenates

To evaluate the metabolic profiles of EH and HV2, the extracted ion chromatogram (EIC) of liver and kidney homogenates incubated for 3 h and 5 min, respectively, with EH or HV2, at 37 °C was obtained (Fig. 3). Eight metabolites of EH and HV2 were found in both the liver and kidney homogenates. Tables 2–3 shows the retention time (min), peptide sequence, molecular weight, and observed *m/z* of parent peptide and metabolites in each tissue.

EH shows a monoisotopic accurate mass of 7291.2388 ($[M + H]^+$) representing the charged states $[M + 4H]^{4+}$, $[M + 5H]^{5+}$, $[M + 6H]^{6+}$, and $[M + 7H]^{7+}$ in the full scan of the MS spectrum, whereas HV2 shows a monoisotopic accurate mass of 6909.0423 ($[M + H]^+$) representing the charged states $[M + 4H]^{4+}$, $[M + 5H]^{5+}$, and $[M + 6H]^{6+}$. The *m/z* of each chromatographic peak is depicted in Supplemental Figs. 1–14. The results were confirmed by mass determination and the ion product identification. The collision-induced dissociation (CID) spectra of EH and HV2 (Fig. 4) indicated that the products were predominantly b and y ions. The CID spectrum of EH (Fig. 4A) shows the C-terminal y fragment ions y_3, y_2, and y_6, and the N-terminal b fragment ions b_2, b_{60}, b_{61}, b_{62}, b_{64}, b_{65}, b_{66}, and b_{67}. Structural assignments of the various EH fragment ions are shown in Fig. 4A. The signal and mass accuracy in the MS^n spectra of the seventh-charged ion (*m/z* 1042.17322) were consistent with the theoretically calculated b- and y- ions of EH. The CID spectrum of HV2 (Fig. 4B) revealed the C-terminal y fragment ions y_6, y_3, and y_2 and the N-terminal b fragment ions b_{58}, b_{59}, b_{61}, b_{62}, b_{63}, and b_{64}. Fragment ions corresponding to PEE (356.144), PEEY (519.207), and PEEYL (632.291) were also observed. Structural assignment of HV2 is shown in Fig. 4B. The CID spectra of all metabolites with structural assignments are shown in Tables 4–5 and Supplemental Figs. 15–30.

After incubation of EH with liver and kidney homogenates, a number of degradation products was observed. As can be seen in Table 2, all EH metabolites were derived by cleavage of the peptide backbone. Proteolytic hydrolysis at the C-terminus and cleavage sites was observed between residues Asn55-Asn56 (M1), Asn56-Gly57 (M2), Gly57-Asp58 (M3), Phe59-Glu60 (M4), Glu64-Glu65 (M5), Glu65-Tyr66 (M6), Tyr66-Leu67 (M7), and Leu67-Gln68 (M8). Metabolites M1 to M5 were only observed in kidney homogenates, M6 and M7 only in liver homogenates, and M8 in both kidney and liver homogenates. However, the M8 concentration in liver homogenates was much higher than that of kidney homogenates.

After incubation of HV2 with liver and kidney homogenates, eight C-terminal-truncated products of HV2 were found: $HV2_{1–52}$ (M9), $HV2_{1–53}$ (M10), $HV2_{1–54}$ (M11), $HV2_{1–56}$ (M12), $HV2_{1–61}$ (M13), $HV2_{1–62}$ (M14), $HV2_{1–63}$ (M15), and $HV2_{1–64}$ (M16), which were similar to those of EH. The metabolic products of EH and HV2 had the same cleavage sites and were all truncated at the C-terminal.

3.1.3. Inhibition of EH and HV2 metabolism in liver and kidney homogenates

To identify the proteolytic enzymes involved in the degradation of EH and HV2, the effects of various peptidase inhibitors on the degradation of EH and HV2 in rat tissue homogenates were evaluated. Comparing Fig. 5A with Fig. 5B, we found that in kidney homogenates, DL-thiorphan inhibited the production of M1-M5 from EH and increased the generation of M7 and M8. The metabolism of EH likely slowed down because of the presence of inhibitors, thus increasing the probability of generating M7 and M8. Similarly, we found DL-thiorphan reduced the generation of M9-M12 and increased the generation of M15 and M16 from HV2 by comparing Fig. 5C with Fig. 5D. M13 was not significantly suppressed by DL-thiorphan. In liver homogenates, AEBSF

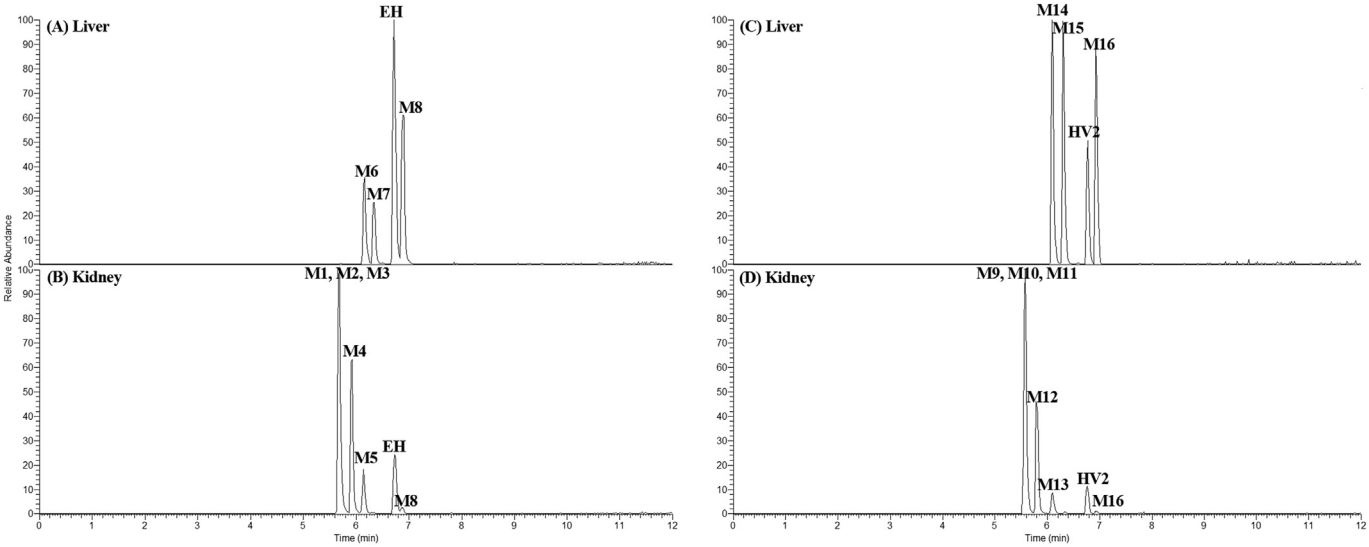


Fig. 3. Extracted ion chromatograms (EICs) from UPLC-ESI-MS analysis of EH, HV2, and sixteen metabolites of incubation in rat liver and kidney homogenates. (A) Representative EICs of EH metabolites formed in rat liver homogenates conducted at 37 °C for 3 h. (B) Representative EICs of EH metabolites formed in rat kidney homogenates conducted at 37 °C for 5 min. (C) Representative EICs of HV2 metabolites formed in rat liver homogenates conducted at 37 °C for 3 h. (D) Representative EICs of HV2 metabolites formed in rat kidney homogenates conducted at 37 °C for 5 min.

Table 2
Major metabolites of EH identified after 3-h incubation in rat liver homogenates and 5-min incubation in rat kidney homogenates.

Analyte	Retention time (min)	Peptide sequence	*m/z* observed				Liver	Kidney
			$[M + 7H]^{7+}$	$[M + 6H]^{6+}$	$[M + 5H]^{5+}$	$[M + 4H]^{4+}$		
EH	6.73	EPRITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDFEEIPEEYLQ	1042.17322	1215.70154	1458.64001	1823.04944	Yes	Yes
EH_{1-68}								
M1	5.67	EPRITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHN	818.64813	954.92151	1145.70435	1431.87769	No	Yes
EH_{1-55}								
M2	5.67	EPRITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNN	834.93964	973.92865	1168.51318	1460.64014	No	Yes
EH_{1-56}								
M3	5.67	EPRITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNG	843.08545	983.43134	1179.91724	1474.64478	No	Yes
EH_{1-57}								
M4	5.90	EPRITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDF	880.52826	1027.11523	1232.33765	1540.42041	No	Yes
EH_{1-59}								
M5	6.16	EPRITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDFEEIPE	965.99567	1126.82739	1352.19324	1689.73828	No	Yes
EH_{1-64}								
M6	6.14	EPRITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDFEEIPEE	984.42981	1148.16687	1377.80078	1721.99805	Yes	No
EH_{1-65}								
M7	6.34	EPRITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDFEEIPEEY	1007.58209	1175.51208	1410.41272	1762.76440	Yes	No
EH_{1-66}								
M8	6.91	EPRITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDFEEIPEEYL	1023.87964	1194.19128	1433.22876	1791.03674	Yes	Yes
EH_{1-67}								

Table 3
Major metabolites of HV2 identified after 3-h incubation in rat liver homogenates and 5-min incubation in rat kidney homogenates.

Analyte	Retention time (min)	Peptide sequence	*m/z* observed			Liver	Kidney
			$[M + 6H]^{6+}$	$[M + 5H]^{5+}$	$[M + 4H]^{4+}$		
HV2	6.78	ITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDFEEIPEEYLQ	1152.00085	1382.20081	1727.49927	Yes	Yes
$HV2_{1-65}$							
M9	5.58	ITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHN	891.22369	1069.26721	1336.58289	No	Yes
$HV2_{1-52}$							
M10	5.58	ITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNN	910.23065	1092.07556	1364.84338	No	Yes
$HV2_{1-53}$							
M11	5.58	ITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNG	919.73358	1103.47949	1379.09912	No	Yes
$HV2_{1-54}$							
M12	5.78	ITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDF	963.41779	1156.10046	1444.62354	No	Yes
$HV2_{1-56}$							
M13	6.12	ITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDFEEIPE	1063.12939	1275.35437	1594.19177	No	Yes
$HV2_{1-61}$							
M14	6.09	ITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDFEEIPEE	1084.63501	1301.16162	1626.20105	Yes	No
$HV2_{1-62}$							
M15	6.30	ITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDFEEIPEEY	1111.81372	1333.97644	1666.96863	Yes	No
$HV2_{1-63}$							
M16	6.92	ITYTDCTESGQNLCLCEGSNVCGKGNKCILGSNGKGNQCVTGEGTPKPESHNNGDFEEIPEEYL	1130.66052	1356.39209	1695.48840	Yes	Yes
$HV2_{1-64}$							

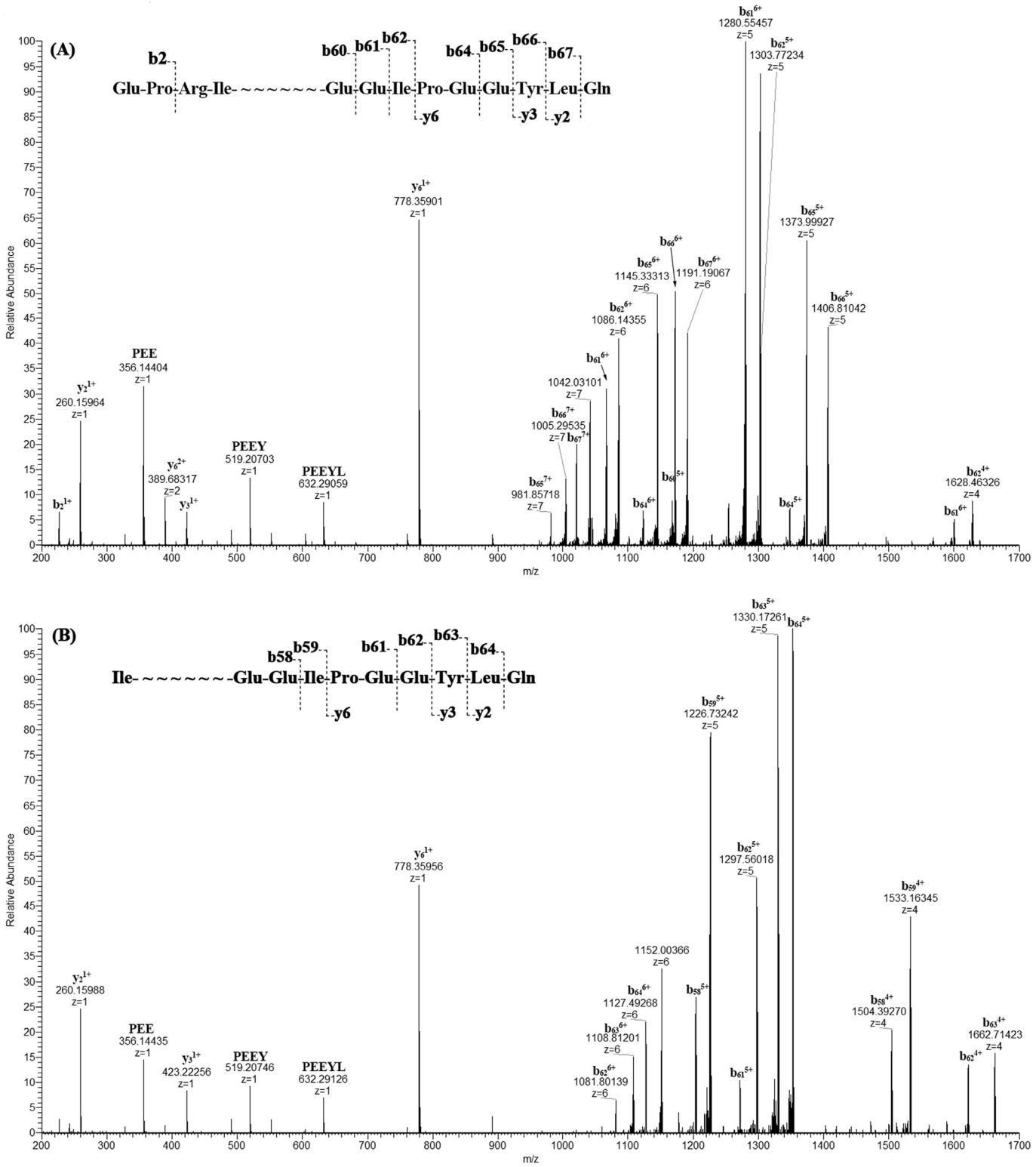


Fig. 4. Collision-induced dissociation (CID) spectra of EH and HV2 (below) with fragmentation pattern indicated (above). Theoretical values of fragment ions (generated from the Prospector software) are depicted in the fragment ion assignments that match the experimental values. (A) MS/MS spectra of EH (charge state +7, *m/z* 1042.17322) using a normalized collision energy of 23 and an isolation width of 3 Da. (B) MS/MS spectra of HV2 (charge state +6, *m/z* 1152.00085) using a normalized collision energy of 26 and an isolation width of 3 Da.

and NEM inhibited the production of M6-M8 from EH and M14-M16 from HV2 (Fig. 6). The inhibition rate of peptidases inhibitors was calculated by integrated semi-quantitative determination (Esposito et al., 2017). The formula for inhibition rate (%) was as follows: (1-peak area of metabolites with inhibitor/peak area of metabolites without inhibitor) × 100%. In kidney homogenates, > 50% inhibition of M1–M5 and M9–M12 production was observed with the selective enkephalinase inhibitor thiorphan; enkephalinase is a type of

Table 4
Identification of EH and metabolites using product ion mass spectrometry data.

Analyte	Characteristic product ions													
EH	227.10194 ($z = 1$, b_2)	260.15964 ($z = 1$,y_2)	356.14404 ($z = 1$, PEE)	423.22244 ($z = 1$,y_3)	519.20703 ($z = 1$, PEEY)	632.29059 ($z = 1$, PEEYL)	778.35901 ($z = 1$,y_6) 389.68317 ($z = 2$,y_6)	1254.74622 ($z = 5$, b_{60})	1067.12903 ($z = 6$, b_{61}) 1280.55457 ($z = 5$, b_{61}) 1600.44141 ($z = 4$, b_{61})	1086.14355 ($z = 6$, b_{62}) 1303.77234 ($z = 5$, b_{62}) 1628.46326 ($z = 4$, b_{62})	1123.82581 ($z = 6$, b_{64}) 1348.39063 ($z = 5$, b_{64})	981.85718 ($z = 7$, b_{65}) 1145.33313 ($z = 6$, b_{65}) 1373.99927 ($z = 5$, b_{65})	1005.29535 ($z = 7$, b_{66}) 1172.51025 ($z = 6$, b_{66}) 1406.81042 ($z = 5$, b_{66})	1021.30756 ($z = 7$, b_{67}) 1191.19067 ($z = 6$, b_{67})
M1	478.27603 ($z = 1$, b_4-H_2O)	579.32324 ($z = 1$, b_5-H_2O)	627.28986 ($z = 2$,y_{12})	742.38593 ($z = 1$,b_6-H_2O)	973.61993 ($z = 5$, y_{48})		993.82874 ($z = 5$, y_{49})	1026.44177 ($z = 5$, y_{50})						
M2	478.27594 ($z = 1$, b_4-H_2O)	579.32330 ($z = 1$, b_5-H_2O)	633.78735 ($z = 2$,y_{12})	742.38593 ($z = 1$,b_6-H_2O)	996.22852 ($z = 5$, y_{49})		1016.63782 ($z = 5$, y_{50})	1049.25049 ($z = 5$, y_{51})						
M3	478.27588 ($z = 1$, b_4-H_2O)	579.32312 ($z = 1$, b_5-H_2O)	662.29779 ($z = 2$,y_{13})	712.82141 ($z = 2$,y_{14})	742.38580 ($z = 1$,b_6-H_2O)		984.82703 ($z = 5$, y_{49})	1007.63251 ($z = 5$, y_{50})	1028.04199 ($z = 5$, y_{51})		1060.65466 ($z = 5$, y_{52})			
M4	579.32324 ($z = 1$, b_5-H_2O)	742.38593 ($z = 1$,b_6-H_2O)	1060.25073 ($z = 5$, y_{52})	1080.66089 ($z = 5$, y_{53})	1113.07324 ($z = 5$, y_{54})		996.60162 ($z = 6$, b_{58}-H_2O)							
M5	245.11266 ($z = 1$, y_2)	1045.78931 ($z = 6$, b_{60}) 1254.74670 ($z = 5$, b_{60})	1067.29626 ($z = 6$, b_{61}) 1280.35437 ($z = 5$, b_{61})	1086.14368 ($z = 6$, b_{62})										
M6	227.10181 ($z = 1$, PE)	356.14395 ($z = 1$, y_3-H_2O)	374.15448 ($z = 1$,y_3)	1045.78894 ($z = 6$, b_{60}) 1254.54541 ($z = 5$, b_{60})	1067.12830 ($z = 6$, b_{61})	1280.55334 ($z = 5$, b_{61})	1085.97595 ($z = 6$, b_{62}) 1303.17017 ($z = 5$, b_{62})							
M7	227.10191 ($z = 1$, PE)	293.11227 ($z = 1$, y_2-H_2O)	356.14410 ($z = 1$, PEE)	537.21747 ($z = 1$,y_4)	1045.62134 ($z = 6$, b_{60}) ($z = 5$, b_{60})	1254.54700	1067.12915 ($z = 6$, b_{61}) 1280.55408 ($z = 5$, b_{61})	1086.14343 ($z = 6$, b_{62}) 1303.17114 ($z = 5$, b_{62})		1123.82629 ($z = 6$, b_{64})	1145.50024 ($z = 6$, b_{65})			
M8	227.10201 ($z = 1$, PE)	295.16449 ($z = 1$, y_2)	356.14420 ($z = 1$, PEE)	650.30127 ($z = 1$,y_5)	1045.62146 ($z = 6$, b_{60}) 1254.54614 ($z = 5$, b_{60})	1067.29578 ($z = 6$, b61) 1280.55493 ($z = 5$, b61) 1600.19104 ($z = 4$, b61)	1086.31030 ($z = 6$, b62) 1303.37219 ($z = 5$, b62)	1124.16028 ($z = 6$, b64) 1348.59131 ($z = 5$, b64)		981.71545 ($z = 7$, b65) 1145.33313 ($z = 6$, b65) 1373.99768 ($z = 5$, b65)	1005.15186 ($z = 7$, b66) ($z = 6$, b66)	1172.34509		

Table 5
Identification of HV2 and metabolites using product ion mass spectrometry data.

Analyte	Characteristic product ions											
HV2	260.15988 ($z = 1, y_2$)	356.14435 ($z = 1$, PEE)	423.22256 ($z = 1, y_3$)	519.20746 ($z = 1$, PEEY)	632.29126 ($z = 1$, PEEYL)	778.35956 ($z = 1, y_6$)	1204.11682 ($z = 5, b_{58}$) 1504.39270 ($z = 4, b_{58}$)	1226.73242 ($z = 5, b_{59}$) 1533.16345 ($z = 4, b_{59}$)	1271.95105 ($z = 5, b_{61}$)	1081.80139 ($z = 6, b_{62}$) 1297.56018 ($z = 5, b_{62}$) 1621.94739 ($z = 4, b_{62}$)	1108.81201 ($z = 6, b_{63}$) 1330.17261 ($z = 5, b_{63}$) 1662.71423 ($z = 4, b_{63}$)	1127.49268 ($z = 6, b_{64}$) 1352.98962 ($z = 5, b_{64}$)
M9	215.13885 ($z = 1, b_2$)	583.24646 ($z = 1, y_5$)	973.82294 ($z = 5, y_{48}$)	993.63104 ($z = 5, y_{49}$)	855.37024 ($z = 6, y_{50}$) 1026.44312 ($z = 5, y_{50}$)	1046.65271 ($z = 5, y_{51}$)						
M10	215.13879 ($z = 1, b_2$)	996.63000 ($z = 5, y_{49}$)	1016.63947 ($z = 5, y_{50}$)	874.54480 ($z = 6, y_{51}$) 1049.25159 ($z = 5, y_{51}$)	1069.46130 ($z = 5, y_{52}$)							
M11	215.13860 ($z = 1, b_2$)	1008.03485 ($z = 5, y_{50}$)	1028.04346 ($z = 5, y_{51}$)	884.04779 ($z = 6, y_{52}$) 1060.65540 ($z = 5, y_{52}$)	1080.86462 ($z = 5, y_{53}$)							
M12	215.13869 ($z = 1, b_2$)	1060.25244 ($z = 5, y_{52}$)	1080.66223 ($z = 5, y_{53}$)	927.89758 ($z = 6, y_{54}$) 1113.07458 ($z = 5, y_{54}$)	1133.28406 ($z = 5, y_{55}$)							
M13	245.11267 ($z = 1$, y2)	1178.30786 ($z = 5$, b57)	1204.11584 ($z = 5$, b58) 1504.89294 ($z = 4$, b58)	1226.93225 ($z = 5$, b59)								
M14	227.10229 ($z = 1$, PE)	356.14490 ($z = 1, y_3$-H_2O)	374.15573 ($z = 1, y_3$)	1203.91980 ($z = 5, b_{58}$) 1504.64478 ($z = 4, b_{58}$)	1226.53540 ($z = 5, b_{59}$)	1271.75415 ($z = 5, b_{61}$)						
M15	227.10229 ($z = 1$, PE)	293.11273 ($z = 1, y_2$-H_2O)	356.14465 ($z = 1$, PEE)	537.21826 ($z = 1, y_4$)	1203.91675 ($z = 5, b_{58}$) 1504.64380 ($z = 4, b_{58}$)	1226.53247 ($z = 5, b_{59}$) 1533.16577 ($z = 4, b_{59}$)	1271.95129 ($z = 5, b_{61}$)	1297.76160 ($z = 5, b_{62}$)				
M16	227.10313 ($z = 1$, PE)	295.16611 ($z = 1, y_2$)	356.14636 ($z = 1$, PEE)	519.21069 ($z = 1$, PEEY)	650.30585 ($z = 1, y_5$)	1203.92603 ($z = 5, b_{58}$) 1504.65247 ($z = 4, b_{58}$)	1226.54236 ($z = 5, b_{59}$) 1532.91443 ($z = 4, b_{59}$)	1271.95313 ($z = 5, b_{61}$)	1081.80188 ($z = 6, b_{62}$) 1297.76819 ($z = 5, b_{62}$)	1108.98621 ($z = 6, b_{63}$) 1330.38123 ($z = 5, b_{63}$)		

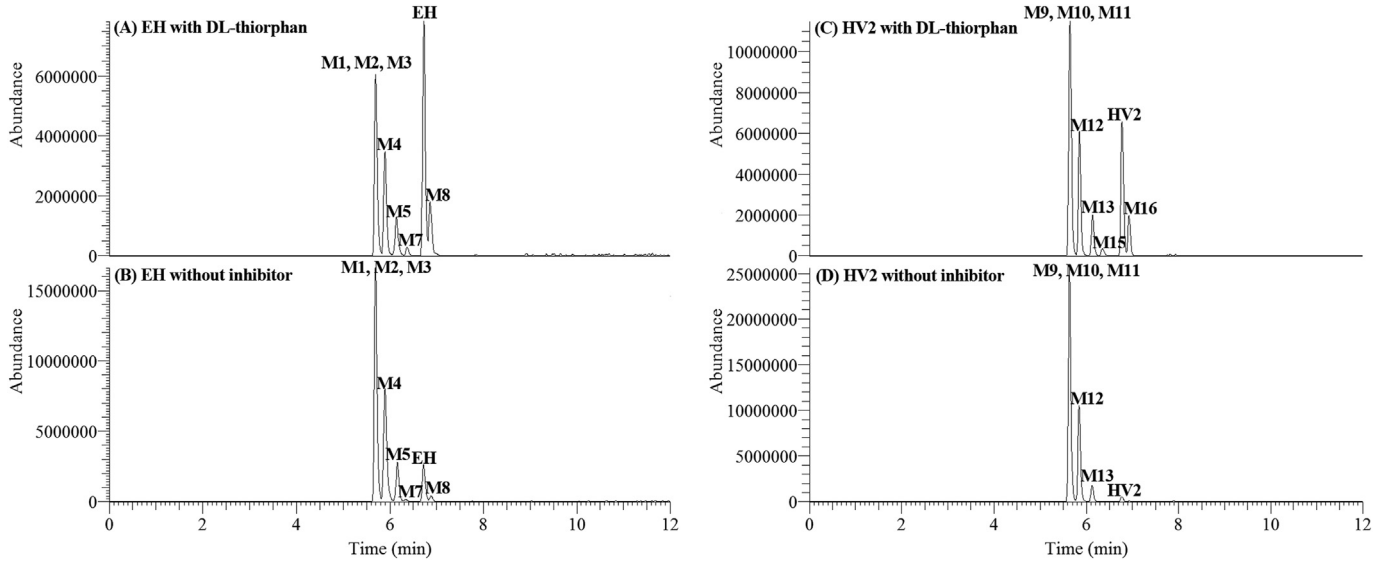


Fig. 5. Extracted ion chromatograms (EICs) of the role of peptidase inhibitors in EH and HV2 metabolism in rat kidney homogenates. (A) EH in rat kidney homogenate pre-incubated with DL-thiorphan for 30 min conducted at 37 °C for 5 min. (B) EH in rat kidney homogenate pre-incubated without peptidase inhibitors conducted at 37 °C for 5 min. (C) HV2 in rat kidney homogenate pre-incubated with DL-thiorphan for 30 min conducted at 37 °C for 5 min. (D) HV2 in rat kidney homogenate pre-incubated without peptidase inhibitors conducted at 37 °C for 5 min.

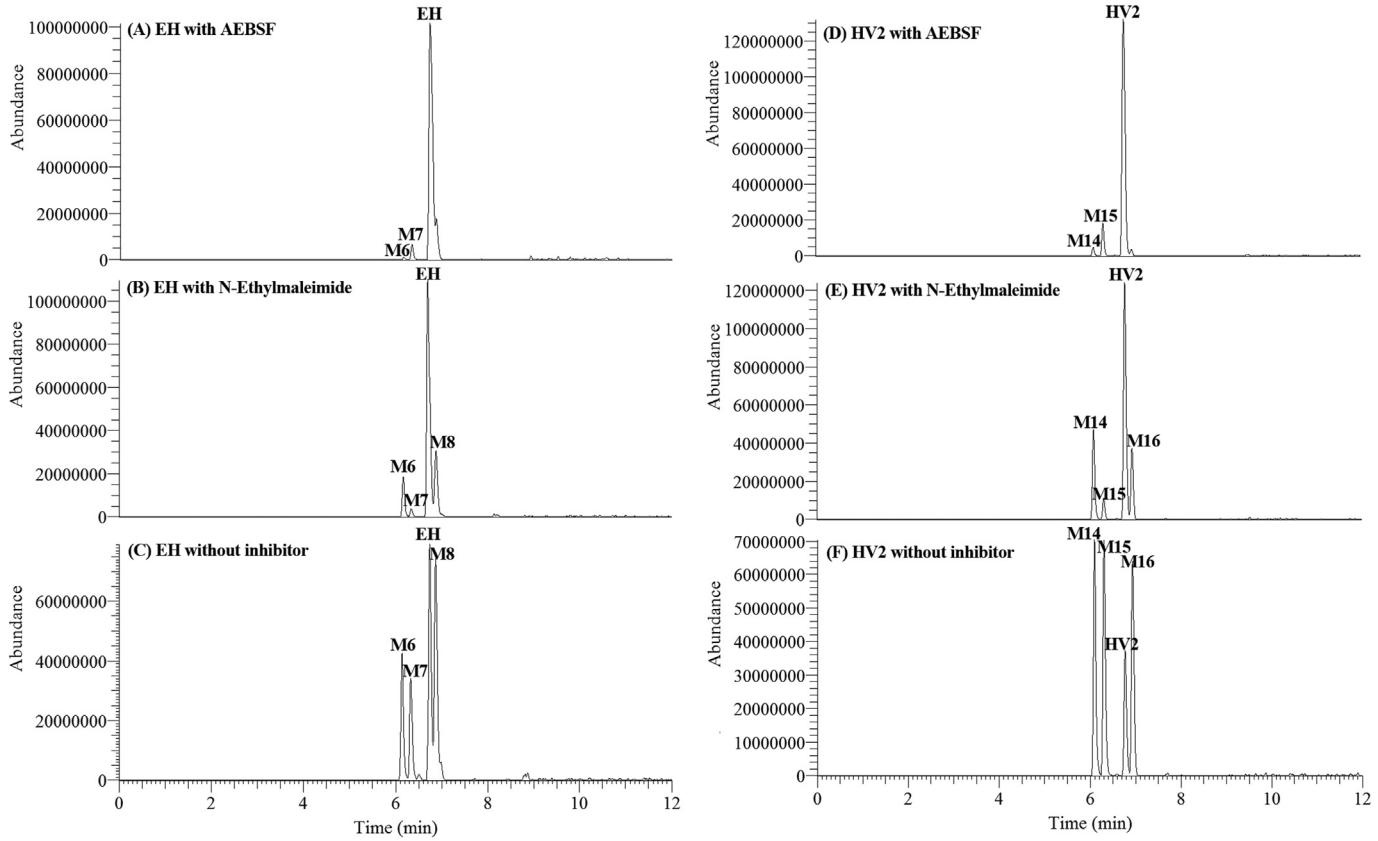


Fig. 6. Extracted ion chromatograms (EICs) of the role of peptidase inhibitors in EH and HV2 metabolism in rat liver homogenates. (A) EH in rat liver homogenate pre-incubated with AEBSF or NEM for 10 min conducted at 37 °C for 3 h. (B) EH in rat liver homogenate pre-incubated without peptidase inhibitors conducted at 37 °C for 3 h. (C) HV2 in rat liver homogenate pre-incubated with AEBSF or NEM for 10 min conducted at 37 °C for 3 h. (D) HV2 in rat liver homogenate pre-incubated without peptidase inhibitors conducted at 37 °C for 5 min.

metallocarboxypeptidase. However, no inhibition of M1–M5 or M9–M13 production was observed with the endopeptidase inhibitor, aminopeptidase inhibitor, serine peptidase inhibitor, and cysteine peptidase inhibitor (data not shown). In liver homogenates, a specific serine peptidase inhibitor of AEBSF had a significant inhibitory effect on the production of M6 and M8 from EH and M14 and M16 from HV2, with 77% or more inhibition. NEM, a cysteine peptidase inhibitor by alkylation of active site thiol, reduced M7 production from EH and M15

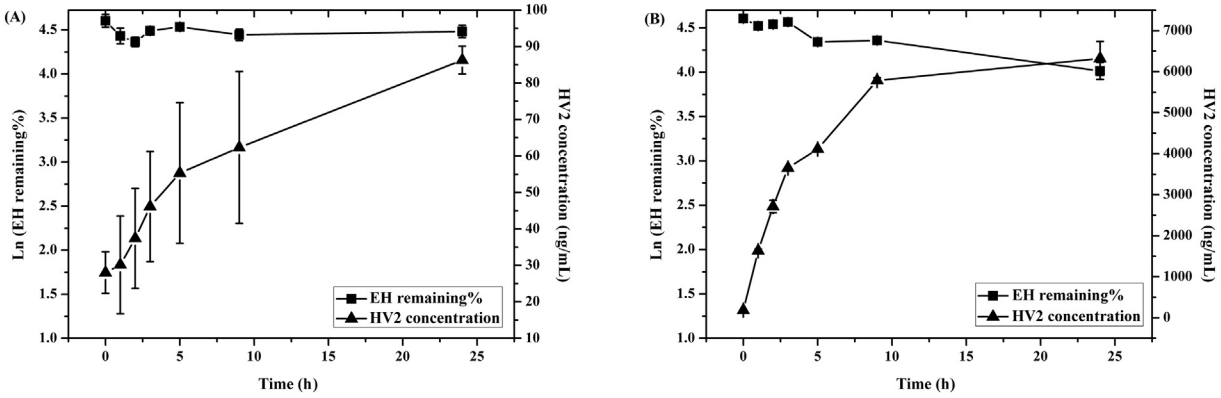


Fig. 7. Time curves of EH cleavage and HV2 production upon EH incubation in presence of FXa and FXIa. The values on the left y axis represent Ln (EH remaining %) and the EH remaining % is the percentage recovery versus T_{0min}. The values on the right y axis represent the concentration (ng/mL) of HV2. Data represent mean ± SD from three replicate incubations. (A) EH incubation with FXa conducted at 37 °C for 24 h (B) EH incubation with FXIa conducted at 37 °C for 24 h.

from HV2 by approximately 90%. However, the metallopeptidase inhibitor thiorphan, phosphoramidon, and bestatin had no inhibitory effect on M6-M8 and M14-M16 production.

3.2. In vitro conversion of EH to HV2

3.2.1. The cleavage of EH to release HV2 by FXa and FXIa

We evaluated the dissociation characteristics of EH in the presence of FXa and FXIa by measuring the remaining EH and the yield of HV2, respectively. The data as shown in Fig. 7 indicated that EH showed a time-dependent release of HV2 in the presence of FXIa. The conversion rate (%) at 24 h was calculated by the following formula: (molarity of HV2 at 24 h/molarity of EH at 0 min) × 100%. In the case of incubation with FXa for 24 h, the remaining EH was 89% and the production of HV2 was 86 ng/mL, in which the conversion rate of EH to HV2 was approximately 0.4%. In the case of incubation with FXIa for 24 h, the remaining EH was 56%, and the production of HV2 was 6323 ng/mL, in which the conversion rate of EH to HV2 was approximately 42%.

3.2.2. Release of HV2 from EH in fresh whole blood and whole-blood clot homogenate

The comparison between EH incubated with fresh whole blood and whole-blood clot homogenates was evaluated by UPLC-MS/MS (Fig. 8). As shown in Fig. 8C and Fig. 8F, no significant endogenous substances interfered with the analysis of EH and HV2 in the blank fresh whole blood and whole-blood clot homogenates. As shown in Fig. 8A and Fig. 8D, EH had been reduced by 20% in whole-blood clot homogenates comparing with that in fresh whole blood after incubating for 3 h. Typical chromatograms of EH in fresh whole blood suggested a small amount of HV2 at this time range. Typical chromatograms of EH in whole-blood clot homogenates demonstrated significantly increased HV2 during this time. HV2 levels were determined by comparing the HV2 response in fresh whole blood with the response in whole-blood clot homogenates. The amount of HV2 produced in whole-blood clot homogenates was > 20 times higher than that in fresh whole blood at 3 h. Therefore, the reaction of EH with whole-blood clot homogenates caused a marked release of HV2.

4. Discussion

Thrombin plays a vital role in thrombosis in patients with cardiovascular diseases (CVDs). The development of antithrombotic drugs with site-directed thrombin inhibition is promising (Lyer and Fareed, 1996; Mehta et al., 2014; Yeh et al., 2016). Hirudin is reportedly the strongest and most specific natural inhibitor of thrombin. EH was developed to deliver active HV2 molecules only to the thrombosis site, at which the levels of FXa and FXIa increase after activation of the coagulation system. The prodrug design was expected to improve

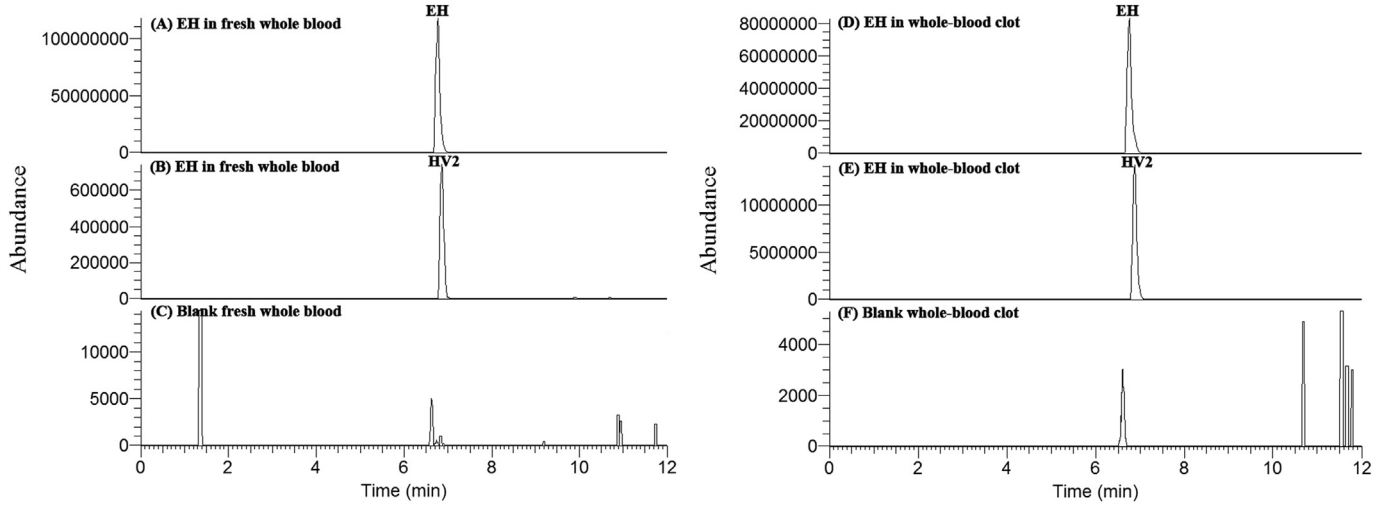


Fig. 8. Extracted ion chromatograms (EICs) of EH transforming to HV2 in fresh whole blood and whole-blood clot homogenates. The comparison between EH in fresh whole blood (A for EH and B for HV2), EH in whole-blood clot homogenates (D for EH and E for HV2), blank fresh whole blood (C), and blank whole-blood clot homogenates (F) provides evidence for the cleavage of EH to release HV2. A typical experiment of 3 h is shown.

availability and reduce the risk of bleeding. The current study evaluated the intended prodrug action by comparing the metabolic characteristics of EH and HV2 and conversion of EH to HV2 in different biological matrices.

We evaluated the metabolic stability of the peptides in different tissue homogenates. EH and HV2 were acutely unstable in kidney homogenates compared with that in liver homogenates, whereas they were stable in blood. The data suggest that the kidney is the main metabolic organ for EH and HV2; this is in agreement with previous reports stating that recombinant hirudins (desirudin and lepirudin) are primarily metabolised and cleared via the kidneys (van Es et al., 2013; Graetz et al., 2011).

Similar metabolites were obtained from EH and HV2 in liver and kidney homogenates, and the metabolic sites of all EH and HV2 metabolites were identical. These results indicate that the addition of the EPR amino acid sequence at the N-terminus of HV2 does not alter the metabolic profile of HV2, which is consistent with the original intention of the EH design. Our results show that EH and HV2 were metabolised in liver and kidney homogenates to metabolites with 1 to 13 amino acids removed from the C-terminus. The C-terminal-truncated products of HV2 are consistent with the reported C-terminal-truncated standards of rHirudin (hirudin variant rHV2-Lys47) produced by enzymatic hydrolysis with carboxypeptidase Y (Maftouh et al., 1991). C-terminal-truncated metabolites have also been reported as the major urinary components after intravenous (i.v.) administration of the recombinant hirudin derivative CX-397 and the MDL 28050 analogue in rats (Knadler et al., 1992; Komatsu and Hayashi, 1999).

The EH and HV2 metabolites in the liver and kidney were distinct, which may be a result of differences in the metabolic enzymes of both organs. The results show that the metallocarboxypeptidases present in the kidney (Beynon and Bond, 1983; Guder and Ross, 1984) are primarily responsible for the metabolism of EH and HV2 in the kidney. Serine and cysteine carboxypeptidases are primarily responsible for the EH and HV2 in the liver metabolism, whereas amino peptidase and endopeptidase inhibitors do not affect EH and HV2 metabolism. Carboxypeptidases are exopeptidases that hydrolyse the C-terminal peptide bonds of peptides and proteins (Rawlings and Salvesen, 2013). It is presumed that M6-M8 from EH and M14-M16 from HV2 are primarily metabolised by serine and cysteine carboxypeptidases in the liver. In addition, in the kidney, EH and HV2 may be primarily metabolised by metallocarboxypeptidases to M1–M5 and M9–M12, respectively.

By the addition of FXa and FXIa recognition sequence, the fusion protein released HV2 in the presence of FXa and/or FXIa. The in vitro cleavage reaction of EH confirmed that FXIa predominantly cleaved EH to release HV2. The recognition sequence of factor XIa is Glu-Pro-Arg (Saporito-Irwin and Van Nostrand, 1995), whereas that of factor Xa is Ile-(Glu/Asp)-Gly-Arg, therefore, factor XIa cleavage of EH is more efficient. Thus, FXIa is predominantly responsible for the cleavage of EH to release HV2. In the present work, whole-blood clot homogenates were used to mimic the activated clotting system. When the coagulation system is activated, the activities of coagulation factors at thrombus are much higher than the activities in the peripheral blood system (Eisenberg et al., 1993). Higher levels of HV2 were produced in whole-blood clot homogenate than those in fresh whole blood, which indicated that EH is transformed into HV2 when the coagulation system is activated; this is dependent on activation of coagulation factors.

The activity of EH and HV2 metabolites was not investigated. Several studies have shown that the C-terminus is crucial for thrombin activation, and loss of C-terminal residues may prevent interaction between hirudin and thrombin (Johnson et al., 1989). The identified metabolites were C-terminal-truncated products of EH and HV2, leading to the speculation that thrombin activation may be affected. The roles of the downstream metabolites of EH and HV2 should be investigated in future pharmacological studies.

In this study, mass spectrometry (MS)-based approaches, which are efficient and reliable for quantification, characterisation, and identification of metabolites, (Katsila et al., 2012; Domon and Aebersold, 2006), were used to elucidate the metabolite structure of EH and HV2 in liver and kidney homogenates. The elucidation of the metabolism of peptide and protein drugs is necessary for improved efficacy and safety (Abid et al., 2009; Sharma et al., 2013; Olsen et al., 2016). However, standardised methodologies to investigate therapeutic biotransformation are still lacking, and clear guidance from regulatory agencies remains to be issued (Hall, 2014). The metabolic profile of recombinant hirudin, which has been first described in this study, provides a reference for biotransformation studies of peptide and protein drugs using UPLC-MS/MS.

5. Conclusion

In summary, our in vitro study revealed the anticipated prodrug characteristics of EH newly designed as an inactive prodrug of hirudin. The metabolic stability profile of EH in blood, liver, and kidney homogenates was consistent with that of HV2, and the metabolic pathways of both EH and HV2 consisted of removal of amino acids from the C-terminus by carboxypeptidases. The conjunction of the EPR peptide with HV2 resulted not only in retention of metabolic characteristics, but also released HV2 via coagulation factors where FXIa is predominant in the activated coagulation system, which is expected to decrease the risk of bleeding. Furthermore, an animal thrombosis model is required to prove EH release of HV2 at highly clot-specific sites in vivo.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ejps.2018.05.025.

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